

NovaBank Predictive Outreach Decision Support

Purpose: Prioritize limited customer-outreach capacity using a transparent, testable ranking model. The public dataset predicts term-deposit campaign response; it is used here as a proxy for retention engagement, not as a direct churn label.

41,188

customer records

11.27%

positive response rate

0.801 → 0.814

AUC: logistic → boosting

0.460 → 0.489

average precision

Recommendation

Approve a 90-day controlled pilot using the gradient boosting model to rank customers by predicted likelihood of positive outreach response. Use a capacity-based rule: contact the top 20% of scored customers, with higher-cost personal calls reserved for the highest-ranked group and lower-cost digital follow-up for the remainder. The model should guide prioritization; it should not automatically determine customer treatment.

Evidence and rationale

- The test set preserves the 11.27% positive-response rate, making ranking metrics and precision-recall analysis more informative than accuracy alone.
- The interpretable logistic regression baseline achieved AUC 0.801 and average precision 0.460. Gradient boosting improved these to AUC 0.814 and average precision 0.489.
- Call duration was excluded because it is known only after a call and would create data leakage. The model therefore uses only information available before outreach.
- The strongest predictive signals include employment conditions, campaign timing, contact channel, prior contacts, and previous campaign outcome. Feature importance indicates predictive contribution, not causation.

Decision rule and scenario interpretation

The top-20% rule is the recommended operating policy because outreach capacity is limited. Threshold analysis is retained as sensitivity evidence: the F2 score peaks near 0.46, while the illustrative cost model reaches its minimum near 0.13 when a false positive is assumed to cost \$15 and a false negative \$500. These cost inputs are assumptions, not verified NovaBank economics; actual value must be established through the pilot.

Pilot design and success measures

Scope	5,000 customers: 2,500 model-selected and 2,500 randomized control
Duration	90 days from first outreach
Primary KPIs	Incremental response lift, cost per successful response, and contact completion rate
Guardrails	Selection rate, precision and recall by age/job group; complaints; workload; model drift

Key risks and controls

- Proxy-target risk: the source outcome is campaign subscription, not churn. Any retention conclusion must be treated as a business hypothesis.
- Fairness risk: compare selection, precision, and recall across age and job groups before scaling; investigate material disparities rather than assuming fairness.
- Drift and operational risk: monitor macroeconomic inputs, outreach fatigue, complaint rates, and capacity; retrain at least every six months or after material performance deterioration.

Submission links: https://github.com/neelimaparupally/Novabank_Analytics_Project/blob/main/novabank_analysis_submission_ready.ipynb

Source: Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems, 62, 22-31. Dataset: UCI Bank Marketing (bank-additional-full.csv). Metrics shown above were cross-checked against the project analysis and generated model-performance figures.